# Blaqks One — Technology Stack

The tools behind the Blaqks One real-estate CRM platform: what each one is, why we use it and the problem it solves.

**In use** running today **At launch** needed for hosting and public sign-up **Later** added with the named feature

## Frontend

### React 18 — In use

**Why:** Widely used component library for building screens from reusable pieces

**Solves:** Keeps a large CRM interface consistent and maintainable

### React Router — In use

**Why:** Standard page navigation for React

**Solves:** Each screen has its own address and survives a reload

### Vite — In use

**Why:** Very fast development server and production build tool

**Solves:** Instant updates while building; small, fast files for users

### JavaScript and CSS — In use

**Why:** Plain, well-known languages with light and dark themes

**Solves:** No heavy framework lock-in; easy to hire for

### JSZip — In use

**Why:** Small library for zip files

**Solves:** Builds Excel workbooks and backups without a large spreadsheet library

## Backend

### Rust — In use

**Why:** Fast, memory-safe language that catches many bugs before release

**Solves:** Reliable server that handles money and permissions safely

### Axum and Tokio — In use

**Why:** Modern Rust web framework and asynchronous runtime

**Solves:** Serves many users at once with little hardware

### SQLx — In use

**Why:** Database library that checks queries and runs migrations

**Solves:** Safe database access and versioned database changes

**Argon2** — In use

**Why:** Recommended password-hashing method

**Solves:** Stolen database copies cannot reveal passwords

**Rustls** — In use

**Why:** Modern, memory-safe encryption library

**Solves:** Encrypted connections without OpenSSL vulnerabilities

## Data and storage

**PostgreSQL 17** — In use

**Why:** Proven, open-source database with strong transactions and integrity rules

**Solves:** One trusted source of truth; each company's data stays separate and consistent

**Docker and Docker Compose** — In use

**Why:** Runs PostgreSQL the same way on every machine

**Solves:** Identical development setup; disposable databases for tests

**Managed PostgreSQL in a UAE region** — At launch

**Why:** Hosted database with automatic backups, updates and failover

**Solves:** No manual database operations; meets UAE data-location expectations

**Object storage in a UAE region (S3-compatible)** — At launch

**Why:** Durable storage for property photos, floor plans and agreement evidence

**Solves:** Media off the database server, cheap to store and served through the CDN

**Valkey (open-source Redis)** — Later

**Why:** In-memory cache and fast counters

**Solves:** Fast public listing pages and real-time features when traffic grows

## Hosting and traffic

**Cloud provider with a UAE region** — At launch

**Why:** For example AWS (UAE) or Azure (UAE North)

**Solves:** Client data stays in or near the UAE under its data-protection law

**Cloudflare CDN** — At launch

**Why:** Global network that serves files and images from near the user

**Solves:** Fast website and photos; blocks attacks and bad traffic before it reaches us

**Cloudflare Turnstile** — At launch

**Why:** Privacy-friendly bot check on sign-up and sign-in

**Solves:** Stops fake sign-ups and automated password guessing

**Nginx (reverse proxy and gateway)** — At launch

**Why:** Industry-standard entry point in front of our services

**Solves:** HTTPS, routing `/api/crm`, `/api/public` and `/api/calendar`, rate limits, security headers

**Forward proxy for outbound calls** — Later

**Why:** Controls which outside services our servers may call

**Solves:** Limits and logs calls to WhatsApp, Meta and AI providers

**Dedicated API gateway (for example Kong)** — Later

**Why:** API keys, usage plans and billing per company

**Solves:** Only needed if Blaqks One offers an API to outside partners

*Private CRM data is never cached by the CDN or proxy; only public files, photos and approved public listings are cached.*

## Accounts, messaging and integrations

**Transactional email service** — At launch

**Why:** For example Amazon SES or Postmark

**Solves:** Sign-up verification, invitations and password recovery reach inboxes

**WhatsApp Business Platform** — Later

**Why:** Official WhatsApp API

**Solves:** Send and receive client messages from the CRM, within Meta's rules

**Meta (Facebook and Instagram) APIs** — Later

**Why:** Official business-account connections

**Solves:** Social messages and ad leads arrive in the CRM

**UAE PASS** — Later

**Why:** UAE government digital identity and signing

**Solves:** Strongest proof for cross-company deal agreements

**Online payments (for example Stripe)** — Later

**Why:** Subscription billing for companies

**Solves:** Companies pay for Blaqks One plans online

**Google Calendar companion** — In use (paused)

**Why:** Separate small Rust service, kept for a later phase

**Solves:** Two-way calendar sync when it resumes

## Quality, delivery and operations

**Node test runner, Cargo test** — In use

**Why:** Unit and API tests for the frontend and backend

**Solves:** Catches broken rules and calculations before release

**Playwright** — In use

**Why:** Real-browser tests of complete user journeys

**Solves:** Proves screens work end to end, on phone and desktop sizes

**Prettier, rustfmt, Clippy, size check** — In use

**Why:** Automatic formatting, code-quality checks and a 150-line file limit

**Solves:** Consistent, readable code that stays easy to change

**Git and GitHub** — In use

**Why:** Version history and code hosting

**Solves:** Every change is tracked and recoverable

**GitHub Actions** — At launch

**Why:** Runs all checks automatically on every change

**Solves:** Nothing is released without passing tests

**Error tracking and uptime monitoring** — At launch

**Why:** For example Sentry plus an uptime checker

**Solves:** We learn about problems before customers report them

**Encrypted backups and restore drills** — At launch

**Why:** Regular, tested database and media backups

**Solves:** Data can be recovered after mistakes or outages

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Agreed with the owner on 15 October 2026. Source: docs/tech-stack.md.